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Leveraging past information and machine learning to accelerate land disturbance monitoring

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ABSTRACT

Near real-time (NRT) monitoring of land disturbances holds great importance for delivering emergency aid, mitigating negative social and ecological impacts, and distributing resources for disaster recovery. Many past NRT techniques were built upon examining the overall change magnitude of a spectral anomaly with a pre-defined threshold, namely the unsupervised approach. However, their lack of fully considering spectral change direction, change date, and pre-disturbance conditions often led to low detection sensitivity and high commission errors, especially when only a few satellite observations were available at the early disturbance stage, eventually resulting in a longer lag to produce a reliable disturbance map. For this study, we developed a novel supervised machine learning approach guided by historical disturbance datasets to accelerate land disturbance monitoring. This new approach consisted of two phases. For the first phase, the supervised approach applied retrospective analysis on historical Harmonized Landsat Sentinel-2 (HLS) datasets from 2015 to 2021, combined with several open disturbance products. The disturbance model was constructed for each condition of consecutive anomaly number, with the aim of enhancing the specificity for delineating early-stage disturbance regions. Then, these stage-based models were applied for an NRT scenario to predict disturbance probabilities with 2022 HLS images incrementally on a weekly basis. To demonstrate the capability of this new approach, we developed an operational NRT system incorporating both the unsupervised and supervised approach. Latency and accuracy were evaluated against 3000 samples that were randomly selected from the five most influential disturbance events of the United States in 2022, based on labels and disturbance dates interpreted from daily PlanetScope images. The evaluation showed that the supervised approach required 15 days (since the start of the disturbance event) to reach the plateau of its F_1 curve (where most disturbance pixels are detected with high confidence), seven days earlier with roughly 0.2 F_1 score improvement compared to the unsupervised approach (0.733 vs. 0.546 F_1 score). Further analysis showed the improvement was mainly due to the substantial decrease in commission errors (17.7% vs 44.4%). The latency component analysis illustrated that the supervised approach only took an average of 4.1 days to yield the first disturbance alert at its fastest daily updating speed, owing to its decreased sensitivity lag. This finding highlighted the importance of using past knowledge and machine learning to reduce detection delays for an NRT monitoring task.

1. Introduction

1.1. Near real-time monitoring

Land disturbance, often defined as any discrete event that occurs outside the natural variability of the land (Zhu et al., 2020), could fundamentally alter land surface composition, condition, and cover types, consequently disrupting ecosystem functioning such as biodiversity (Martínez-Ramos et al., 2016), productivity (Peters et al., 2013), carbon storage (Liang et al., 2014; Seidl et al., 2014). Land disturbances include natural hazards such as wildfire, flooding, tornado, and a variety of land-use shifts led by anthropogenic activities (Thornton et al., 2024; Zhu et al., 2020). There is a pressing need for mapping land disturbance events in a timely manner, which will hold great relevance for mitigating their social and ecological impacts (Verbesselt et al., 2012) and delivering emergency aid to protect lives and infrastructure (Field et al., 2012).

A large-scale Near Real-Time (NRT) monitoring of land disturbances

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has been long limited by a lack of timely satellite data acquisition until multiple global datasets became publicly available with reduced data latency and improved delivery service (Woodcock et al., 2008; Wulder et al., 2018). Earlier satellite-based NRT works were mainly based on coarse-resolution optical images such as the MODIS products, DETER project (Shimabukuro et al., 2006), the BFAST-monitor tool (Verbesselt et al., 2012), and later extended to the medium-resolution dataset owning to the free access policy of Landsat datasets, such as the Global Land Analysis and Discovery (GLAD) alerting system (Hansen et al., 2016), the Geobosques platform (Vargas et al., 2019) and Continuous Change Detection and Classification (CCDC)-adapted approaches (Pasquarella et al., 2017; Ye et al., 2021b). The recent NRT studies incorporate the Synthetic Aperture Radar (SAR) dataset (Bullock et al., 2022; Doblas et al., 2022; Eckerstorfer et al., 2019; Martinis et al., 2018) and 3-m daily PlanetScope data (Francini et al., 2020), further reducing the observation lag by incorporating more frequent acquisition of satellite observations. On the other hand, some advanced approaches combined multiple sensors to shorten the time interval for collecting temporal observations (Reiche et al., 2018; Shang et al., 2022; Tang et al., 2023). For example, Shang et al. (2022) used the Harmonized Landsat Sentinel-2 (HLS) dataset which combines four satellite sensors (Landsat 8, Landsat 9, Sentinel-2A, and Sentinel-2B) and successfully decreased the confirmation latency of a land disturbance to 35 days.

Despite various input dataset types, NRT disturbance algorithms are generally categorized into two broad groups: the cover-based and the anomaly-based approaches. The cover-based approach was built on the land cover classification and cover category change (or called land cover conversion) (Andela et al., 2022; Diniz et al., 2015; Francini et al., 2020; Giglio et al., 2009; Hansen et al., 2016; Reiche et al., 2018, 2021; Vargas et al., 2019). This approach, typically the Global Land Analysis & Discovery (GLAD) Alert system (Hansen et al., 2016), constructs a land-cover classification model based on historical or empirically selected samples, then applies the model to classify each newly collected image and to detect new disturbance pixels if these pixels have been consistently classified as altered land cover categories. The recent emerging studies for NRT land cover mapping (Brown et al., 2022; Yu et al., 2022) could be potentially leveraged for cover-based land disturbance monitoring as well. The cover-based approach often requires an additional step for reducing data noise and normalizing input images (Reiche et al., 2018, 2021) or simply assumes no seasonality for their targeted cover (Hansen et al., 2016). If a disturbance only induces within-type surface change (or named as cover condition change), such as insect disturbance and drought, the cover-based approach might fail to alert the disturbance. Some advanced cover-based methods generated soft classification outputs as a proxy, such as cover probabilities (Reiche et al., 2018, 2021) or cover fractions (Vargas et al., 2019), with an aim to address subtle change as a "continuous" variable (e.g., forest loss). Nevertheless, these methods were only successful in a relatively simple scenario for a two-category classification scheme (e.g., forest vs. non-forest) so far.

The anomaly-based approach detects early disturbance signals by discriminating recent spectral anomalies from time-series dataset (Olsson et al., 2016; Pasquarella et al., 2017; Shang et al., 2022; Tang et al., 2019; Verbesselt et al., 2012; Ye et al., 2021a). The anomaly-based approach was often achieved without the necessity of tackling land cover information. This approach first builds a spectral baseline model by fitting historical time-series datasets from a stable period, and then continuously examines if the newly collected satellite observation reaches a spectral difference over a predefined threshold against its predicted reflectance. The new disturbance event will be confirmed if the minimum number of consecutive anomalies (namely "peek window") is satisfied. The anomaly-based approach could exclude such noises as data seasonality and other natural variations by incorporating cyclic trends and long-term trends into the baseline model. Also, benefiting from directly detecting change magnitudes, the anomaly-based approach is capable of detecting subtle disturbances that do not directly alter land cover types (Yang et al., 2022; Ye et al., 2021b).

Most cover-based or anomaly-based techniques require collecting a minimum number of consecutive observations (*conse*) for disturbance confirmation (e.g., *conse* = 6 in most CCDC-like approaches), as the temporal stability of a disturbance signal aids its differentiation from noisy signals. The setting of *conse* is practically controlled by the user through algorithm configuration (Bullock et al., 2022). Arguably, it is common to employ less-than-required satellite observations for early disturbance alerting (Shang et al., 2022; Ye et al., 2021a), yet fewer post-disturbance observations for decision-making come at the cost of substantially increased commission errors. For example, Bullock et al. (2022) showcased over 80% commission errors in the initial 50 days since the disturbance occurrence date based on the sample-based agreement; likewise, Shang et al. (2022) reported over 80% commission errors when only one or two anomaly observations were identified. This was because these techniques discriminated disturbance signals only relying on their relatively higher overall change magnitudes. When the post-disturbance data collection is short, the anomalies are more likely attributed to ephemeral and noisy signals (e.g., phenological variation, short-term weather extremes, and even data noise), thereby compromising the detection accuracy. Consequently, a long collection of post-disturbance observations is still a must for most operational management tasks, if an NRT disturbance map with an acceptable commission rate is expected.

This study aims to develop a new supervised anomaly-based approach to reduce detection latency and at the same time enhance the detection accuracy. The new approach was inspired by Retrospective Chart Review (RCR), a popular type of medical research design in which pre-recorded, patient-centered datasets are used to examine clinical characteristics (Gill and Kaplan, 2021; Kaji et al., 2014; Vassar and Matthew, 2013). One notable advantage of RCR is that it is easier to access conditions where there is a long latency from the initial exposure and the patient's diagnosis (Hess, 2004) so that the early symptoms could be retrospectively identified. The valuable information gathered from RCR could be further applied to guide subsequent prospective studies. Similarly, the recent availability of numerous land cover datasets and disturbance products (Chuvieco et al., 2019; Johnson and Wittwer, 2008; Latifovic et al., 2016; Rollins, 2009) as well as freely accessed historical satellite images have enabled us to gain insights into early disturbance signals of a disturbance event whose occurrence might be confirmed a lot later. Specifically, we are allowed to design methods to effectively extract its early-stage spectral and temporal features from historical time series, which can be further leveraged to construct machine learning models. With enhanced model specificity and better capability to differentiate noise guided by well-recorded disturbance event information, it is expected that retrospective disturbance analysis will improve both timeliness and accuracy against the current anomaly-based approaches dominated by empirical thresholding. To our knowledge, historical datasets and analyses have not been systematically explored yet for the NRT disturbance mapping tasks. It is still unclear what disturbance predictors are extracted from satellite time series and how disturbance models are constructed for monitoring a variety of land disturbances at their different lag stages.

1.2. Lag components

For this study, the monitoring latency is viewed as an intricacy impacted by five lag components, i.e., observation, data, sensitivity, confirmation, and production lag, from input, algorithm, and production perspectives (Fig. 1). Assessing different proportions of lag components can provide potential treatments for accelerating disturbance alerting. Note that compared to Fig. 1 of Bullock et al. (2022), the assessments of sensitivity and production lags are newly proposed for this study.

The input latency is mainly affected by observation and data lags. An observation lag refers to the time interval between the disturbance occurrence and when the sensor collects the first clear observation,

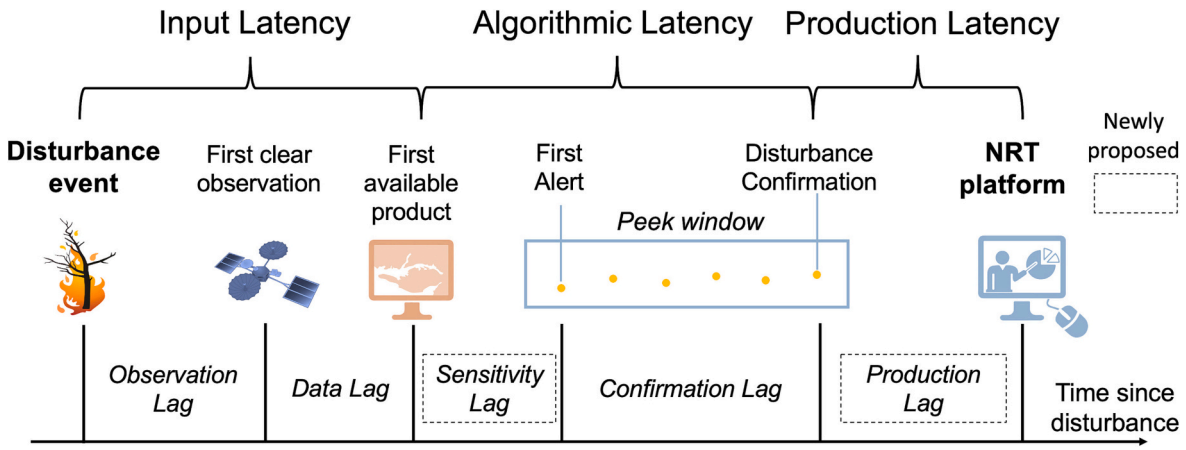


Fig. 1. Graphic explanations for the actual latency from an operational NRT disturbance task which is controlled by five lag components, i.e., observation, data, sensitivity, confirmation, and production lag. Sensitivity and confirmation lag are newly proposed for this study, compared to the work done by Bullock et al. (2022).

which is often related to a satellite revisiting cycle and weather conditions; a data lag is the delay between the first clear observation collection and this observation being preprocessed to a standard reflectance product ready for analytics, which is often systematic and shorter than one week (Bullock et al., 2022).

Algorithmic latency includes sensitivity and confirmation lags. A sensitivity lag refers to the day interval between the date for the first available data product and the date for an algorithm first alerting this disturbance. A sensitivity lag occurs where the disturbance signals are spectrally unobvious at the first observations (e.g., insect disturbances, selective logging), ranging from zero day (i.e., the quickest alert) for those dramatic disturbances, to infinity when the algorithm or the sensor is incapable of detecting the signal. A confirmation lag, also named "algorithm latency" by Bullock et al. (2022), is a lag between the first alert and the disturbance confirmation.

Lastly, a production lag is the operational delay caused by map production and delivery, closely linked to the platform updating interval. The production lag was not well discussed yet in past studies. For each processing, an NRT platform will ingest the new images released, map the latest disturbance, and publish the NRT product. These steps are computationally expensive and time-consuming. Particularly for a large-area monitoring system, it is often practically unfeasible to make a daily update, resulting in a production lag brought by the pre-defined updating day interval.

The remainder of the paper will be organized as follows: we first introduce the study area and dataset used for this research (Section 2), then describe the design of an operational NRT system (Section 3); we will thoroughly compare the latency and accuracy of the system adopting supervised against the unsupervised anomaly-based approach (Section 4.1), report their lag component assessment (Section 4.2), and exhibit distinction of latency curves and lag components from different disturbance events (Section 4.3); finally, we will discuss the comparative performance against other existing NRT methods as well as the limitations (Section 5).

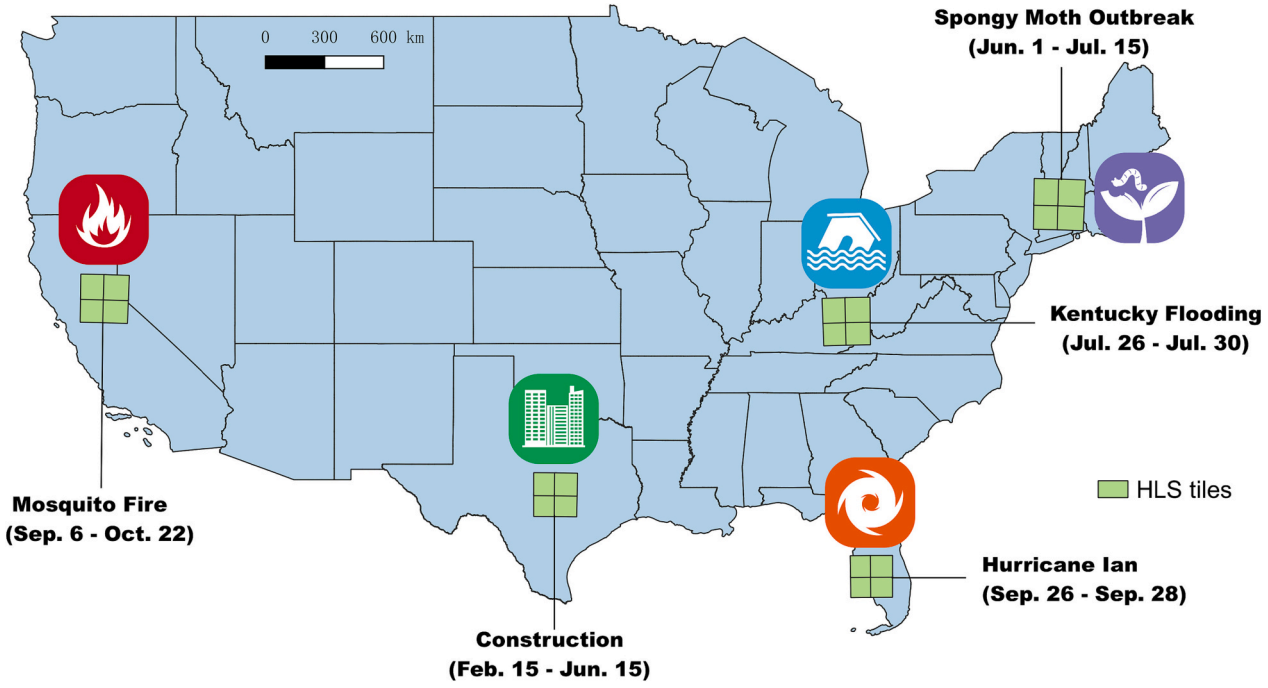


Fig. 2. Five selected sites for NRT monitoring of land disturbance events for 2022 in this study. For each event, four adjacent HLS tiles were used for the test.

2. Study area and dataset

2.1. Study area

We chose the five most influential disturbance events across the Conterminous United States (CONUS) in 2022, including three natural disasters, one forest insect infestation, and one human-induced land surface modification (Fig. 2).

Mosquito fire:

Mosquito fire was the largest wildfire in California occurring in 2022. The massive fire began on September 6 and was 90% contained on October 4, burning a total area of 31,075 ha, mostly in forested regions of El Dorado and Placer counties; more than half of the fire regions (58%) experienced low soil burning severity (Teater, 2023). A total of 11,117 people were evacuated, and 5848 houses were threatened or damaged (Teater, 2023).

Spongy moth:

The spongy moth, formerly known as the gypsy moth, has been a major forest disturbance type in New England since 2015 (Pasquarella et al., 2017). Sponge moth caterpillars hatch from egg mass in early May, begin feeding on the leaves of hardwood trees since early June, and reach peak defoliation by late June or early July as the larvae mature (Pasquarella et al., 2017). Sponge moth defoliation is an ephemeral and cyclic disturbance that occurs repeatedly every year in New England: trees often re-foliate from spongy moth attacks in the summer, although moth-induced repeated defoliation combined with other stressors may lead to a long-term decline in forest health.

Hurricane Ian:

Hurricane Ian, which happened in September of 2022, was the deadliest hurricane to strike Florida since 1935. It peaked as a Category 5 hurricane with 260 km/h winds on September 28 when attacking the west coast of Florida (Karimiziarani and Moradkhani, 2023), causing major inland flooding and landfall in counties such as Lee and Charlotte. The hurricane killed 149 people across Florida and produced catastrophic damage with an estimated loss of 113 billion (NOAA National Centers, 2023), including extensive tree and building damage due to strong wind and floods.

Kentucky flooding:

Between July 26 and July 30, 2022, due to sustained record-breaking rainfall, several devastating floods hit 15 counties in East Kentucky, such as Hidman, Perry, and Prestonsburg counties. The floods resulted in the death of 45 people and displaced thousands more. Entire homes and some communities were swept away by flood water, causing extensive damage to thousands of buildings, vehicles, and other infrastructures in the flooded region, with an estimated cost of over 1.5 billion dollars (NOAA National Centers, 2023).

Construction:

Texas has been one of the hotspots for construction projects in the past decade. Due to the lower cost of living and taxes, Texas, particularly Austin and Houston, has become a big draw to businesses and companies relocating from California and elsewhere. For example, Samsung started construction on a semiconductor manufacturing campus in the suburbs of Austin in 2022. Austin City ranked No.1 in the United States for the housing demand in 2022 with numerous construction sites for real estate developments (Salviati and Warnock, 2023), thereby being selected as the testing site for studying construction events.

2.2. Datasets

Harmonized Landsat Sentinel-2 (HLS) dataset combines four satellite sensors (Landsat 8, Landsat 9, Sentinel-2A, and Sentinel-2B satellites), providing the highest temporal resolution of 2–3 days (Claverie et al., 2018) among all medium-resolution (10–30 m) satellite datasets. Notably, for the latest version of the HLS dataset (version 2.0), the Land Processes Distributed Active Archive Center approximately takes only two days of data lag to release the surface reflectance products, with greatly reduced data lag. Therefore, the HLS datasets were selected as the time series inputs for this study. Each HLS tile has 3660 rows and 3660 columns with 30 m of pixel spacing in both directions, provided over projected map coordinates aligned with the Military Grid Reference System (MGRS). For each disturbance event, we chose four adjacent HLS tiles (2 by 2) for the NRT monitoring and the accuracy assessments (Table 1), and a total of 20 tiles were used for the study. For each tile, we downloaded all harmonized surface reflectance products from 2015 to 2021 for the retrospective analysis; during the later NRT monitoring phase, we downloaded the HLS images of 2022 in an increment of one week which is the predefined production interval for our test.

Besides, we collected four open-access annual CONUS disturbance products from 2016 to 2020, which were used for labeling historical anomalies. They are 1) land cover transition map generated by differencing Land Change Monitoring, Assessment, and Projection (LCMAP) Primary Land Cover maps (Xian et al., 2022) between two neighbor years; 2) LANDFIRE Disturbance Products (Ryan and Opperman, 2013); 3) Landsat Collection 1 Level-3 Burned Area Science Product (Hawbaker et al., 2020); and 4) Aerial Detection Survey (ADS) insect activity dataset which records polygonal forest health information generated by sketch mapping (Johnson and Wittwer, 2008). All these products, except for ADS insect dataset, are originally available at 30-m resolution. We rasterized the ADS dataset to 30-m binary maps (insect vs no-insect regions) to keep aligned with the HLS satellite images.

The validation pixels were sampled by applying a stratified sampling approach on the accumulated NRT land disturbance maps generated by the supervised approach. The accumulated NRT maps were made from merging all NRT detection maps within the assessment window for each disturbance; the assessment window is defined as the disturbance period with an extension of 15 days after a disturbance ends to take account of most delayed detections. We randomly sampled 300 pixels from the "disturbance" and "no disturbance" regions, respectively, in each accumulated NRT map (600 samples for an event), forming 3000 samples in total for five disturbance events. For the response design, three experienced remote sensing experts interpreted the reference disturbance date for each sample pixel as the first date for their identifying visible disturbance signals from Planet Explorer (https://www.planet.com/explorer/). The Planet Explorer provides free access to 3-m daily Planet images. As such, we were capable of determining the actual disturbance date for the reference sample pixel, which was more accurate than interpreting the first anomaly date from the time series (Shang et al., 2022; Ye et al., 2021a, 2023) or using the date right in-between the first anomaly and the last pre-disturbance observation date (Bullock et al., 2022; Reiche et al., 2018). While the occasional cloud/shadow contamination or missing datasets in PlanetScope still might cause an inaccurate interpretation of the exact disturbance date, such impacts were negligible because 1) PlanetScope images provided the

Table 1
Location, disturbance periods, and selected HLS tiles for the disturbance events to be tested for this study.

Disturbance Event	Impacted Region	HLS Tile IDs	Disturbance Period (2022)
Mosquito fire	Central California	10SFH, 10SFJ, 10SGH, 10SGJ	Sep. 6 – Oct. 22
Spongy moth	New England	18TXM, 18TXN, 18TYM, 18TYN	Jun. 1 – Jul. 15
Hurricane Ian	Southwest Florida	17RLK, 17RLL, 17RMK, 17RML	Sep. 26 – Sep. 28
Kentucky flood	Eastern Kentucky	17SKB, 17SKC, 17SLB, 17SLC	Jul. 26 – Jul. 30
Construction	Central Texas	14RNU, 14RNV, 14RPU, 14RPV	Feb. 15 – Jun. 15

densest optical images as daily observations; 2) the beginning dates for some disturbance events (such as Hurricane Ian and Mosquito Fire) were well recorded and ascertained so that the recorded disturbance dates could be directly adopted. The labeling process with PlanetScope images was showcased in Section S1 of the supplementary material.

3. Methodology

A two-phase workflow was designed for this new approach (Fig. 3). For the first phase "retrospective disturbance analysis", we applied an "over-detection" for time-series breakpoints from historical time series to generate a comprehensive sample pool respectively for disturbance or non-disturbance anomalies, under the guidance of the disturbance products; then we developed a specific sample set and built a machine-learning model for each disturbance stage. For the second phase "Near Real-time Monitoring", we implemented a NRT disturbance mapping based on these stage-based disturbance models incrementally with a step of one week for the year 2022, i.e., the system updating interval was one week.

3.1. Phase 1: retrospective disturbance analysis

3.1.1. Anomaly generation

We first applied time-series-based breakpoint detection to generate anomalies from historical HLS datasets (2015–2021) on a per-tile basis. For each HLS tile to be monitored, Stochastic Continuous Change Detection (S-CCD) (Ye et al., 2021a), a near real-time adaptation of Continuous Change Detection and Classification (CCDC) (Zhu and Woodcock, 2014), was applied. S-CCD incorporates the Kalman filter into the CCDC algorithm and recursively updates the anomaly records (see Table S1 in the supplementary), instead of reconstructing the model from scratch when new observations are available (Ye et al., 2021a). The short-memory nature of S-CCD saves disk space from removing the historical images and speeds up the computation by skipping the step of refitting a time-series model, thereby improving model efficiency, especially for large-area NRT monitoring. An improvement we made on Ye et al. (2021a) is to let S-CCD produce spectral anomalies, besides structural breakpoints, for the processing of historical datasets. Spectral anomalies, for this study, refer to the spectral breakpoints within a time series produced by break over-detection. They are related to either real disturbance signals or non-disturbance factors (e.g., phenology shifts and ecosystem recovery). There are two key parameters for controlling the break detection sensitivity for CCDC-like algorithms: change probability threshold (*chg_t*) defines the minimum change magnitude to identify a single breakpoint (the default *chg_t* is 0.99), and *conse* is the minimum number of temporally consecutive anomalies required to confirm a breakpoint (the default *conse* is 6). We used a lower change probability threshold of 0.90 and a shorter *conse* as three observations, ensuring the inclusion of the most subtle disturbance pixels (Cohen et al., 2017) as well as a variety of non-disturbance anomalies. Different from the normal CCDC's way of handling breakpoints, the S-CCD algorithm will skip the model initialization but only update the current anomaly record (see Table S1), considering that the Kalman filter can self-adjust model coefficients for local fluctuations (Ye et al., 2021a).

Structural breakpoints refer to another type of breakpoints that leads to an obvious structural change of the time series, where it is essential to refit a new harmonic curve by using new observations of at least one year to guarantee the model predictability (Zhu et al., 2020). We adopted a more conservative parameter set to generate structural breakpoints, *chg_t* = 0.9999, *conse* = 8. When the last observation of a historical time series was processed, the anomaly records, which represented the current spectral status, were saved out to the disk for the second phase (Fig. 3).

3.1.2. Anomaly labeling

We combined two sources of disturbance maps, respectively from anomaly generation and historical disturbance products, to develop high-confidence spectral anomaly pools. An anomaly-based disturbance map was produced from anomaly records based on the disturbance-extraction strategy proposed in Section 3.3.6 by Zhu et al. (2020). This post-processing excluded "greener" breakpoints identified as increased NIR, decreased Red and SWIR1 bands:

$$\Delta\text{Red} < 0.02 \& \Delta\text{NIR} > -0.02 \& \Delta\text{SWIR1} < 0.02 \quad (1)$$

Where ΔRed, ΔNIR, and ΔSWIR1 are the spectral change magnitudes of the RED, NIR, and SWIR1 bands for the breakpoint. On the other

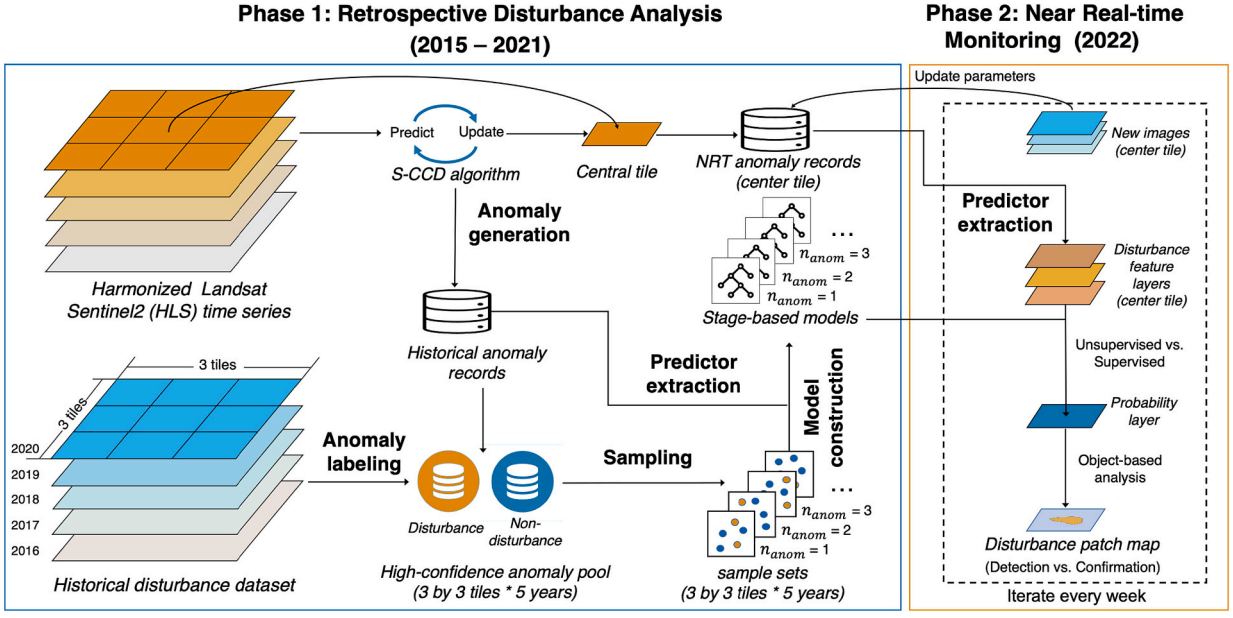


Fig. 3. A two-phase workflow of the supervised near real-time monitoring approach proposed for this study. For Phase 1, eight stage-based models were built for each center tile using anomaly samples from a 3 by 3 moving window of neighbor tiles; the samples were labeled by combining historical disturbance datasets and breakpoint detection from the HLS time series (2015–2021). For Phase 2, the NRT monitoring was implemented using the stage-based retrospective models incrementally for each week of 2022.

hand, we generated an annual product-based disturbance potential map (2016–2020) by fusing four previously mentioned open disturbance products: the pixels were labeled as "disturbance" if this pixel fell within the disturbance regions in at least one open disturbance product. We extracted the high-confidence disturbance pixels from the overlapped disturbance regions between anomaly-based and product-based maps into a disturbance anomaly pool. Those pixels labeled as "disturbance" only in anomaly-based maps were spectral anomalies not covered by historical disturbance regions, thereby representing noisy signals, so we sort them into a non-disturbance anomaly pool.

For each HLS tile, we generated and labeled spectral anomalies on an annual basis, and then combined labeled spectral anomalies from a 3 by 3 HLS tile window centered at the targeted tile (Brown et al., 2019) with a temporal length of the middle five years (2016–2020, see "Phase 1" in Fig. 3). We excluded the year 2015 and 2021 as the S-CCD might suffer inaccuracies for detecting breakpoints at the starting and the ending of a time series. There are two primary motivations for the 3 by 3 spatio-temporal tile window design: 1) the machine learning model could be developed with a better generalization for dealing with disturbance variability using sample acquisition from a larger spatial extent and a longer disturbance history; 2) 3 by 3 tile window considerably decreases variation of local models between two neighborhood tiles, avoiding artifacts on the boundaries of the two tiles (Brown et al., 2019).

3.1.3. Predictor extraction

We extracted 17 disturbance-related predictors from the S-CCD model outputs (not from the disturbance products), which could be categorized into the disturbance-process and the pre-disturbance group. Fig. 4 illustrates these disturbance-related predictors using a sample time series. Six HLS spectral bands, i.e., blue, green, red, NIR, SWIR1, and SWIR2 bands, were considered. We retained the three predictors used in the COntinuous monitoring of Land Disturbance (COLD) algorithm (Zhu et al., 2020) for the disturbance-related breakpoint decision, i.e., 1) multispectral change magnitudes to take account of spectral change direction, 2) normalized change intensity, and 3) spectral consistency angle. Besides, we also included 4) the disturbance date to incorporate seasonality, 5) the span of anomaly days for data temporal density, and 6) multispectral pre-change reflectance to depict land-cover spectral features prior to spectral anomalies being detected. The formulas for these predictors are given in Table 2.

3.1.4. Model construction

To enhance model specificity for different disturbance stages, we individually established one sample set and one disturbance model for each consecutive anomaly number condition (n_{anom} = 1, 2, 3, …, 8, Fig. 5). Zhu et al. (2016) suggested a total of 20,000 samples for the Landsat-based classification, and an HLS image (3660 * 3660) occupies only a half of the total pixel number of the Landsat scene (5000 * 5000), so a total number of 10,000 training samples was selected to build a series of stage-based models for each tile. To evenly distribute these 10,000 samples across the sample years, we allocated 2000 samples for each tile in each year, so for a center tile, a total of 90,000 samples (9 tiles * 5 years * 2000 samples) were used to build each stage-based model (see Fig. 3). A proportional random sampling was chosen because it has shown superiority over a balanced sampling for a classification scheme (Brown et al., 2019; Zhu et al., 2016). However, there are far fewer disturbance anomalies compared to the non-disturbance anomalies in practice, causing the disturbance category to be easily under-represented. The trickier is that the omission and commission error rates are often both increased at a similar degree for the early disturbance stage due to too many noisy signals. But we expected to suppress omission errors so that the potentially disturbed regions were comprehensively provided for the possible subsequent high-resolution damage mapping. As such, we designed a linear function with a negative slope to control disturbance sample proportion, which enabled assigning a higher sample proportion to the disturbance category for the shorter n_{anom} (i.e., the earlier disturbance stage). For each n_{anom}, the sample proportions are denoted as:

$$P_{dist}(n_{anom}) = P_{base} + (conse - n_{anom}) * slope \quad (2)$$

$$P_{nondist}(n_{anom}) = 1 - P_{dist}(n_{anom}) \quad (3)$$

Where $P_{dist}(n_{anom})$ and $P_{nondist}(n_{anom})$ are the proportions of disturbance and non-disturbance categories for each n_{anom}; *conse* is the constant, the number of consecutive anomaly observations to detect a structural breakpoint (*conse* = 8); P_{base} is the baseline proportion when the n_{anom} reaches the maximum, and the parameter *slope* controls decreasing impacts of n_{anom} on P_{dist}. The parameter sensitivity test (see Section S3 of the supplementary material) revealed that P_{base} = 0.1 and *slope* = 0.02 was the only group that keeps an acceptable overall accuracy while maintaining a low omission rate across all n_{anom}.

For each n_{anom} (i.e., 1, 2, 3, …, 8), we trained a random-forest model (*ntrees* = 100) for the binary classification scheme (disturbance vs. no

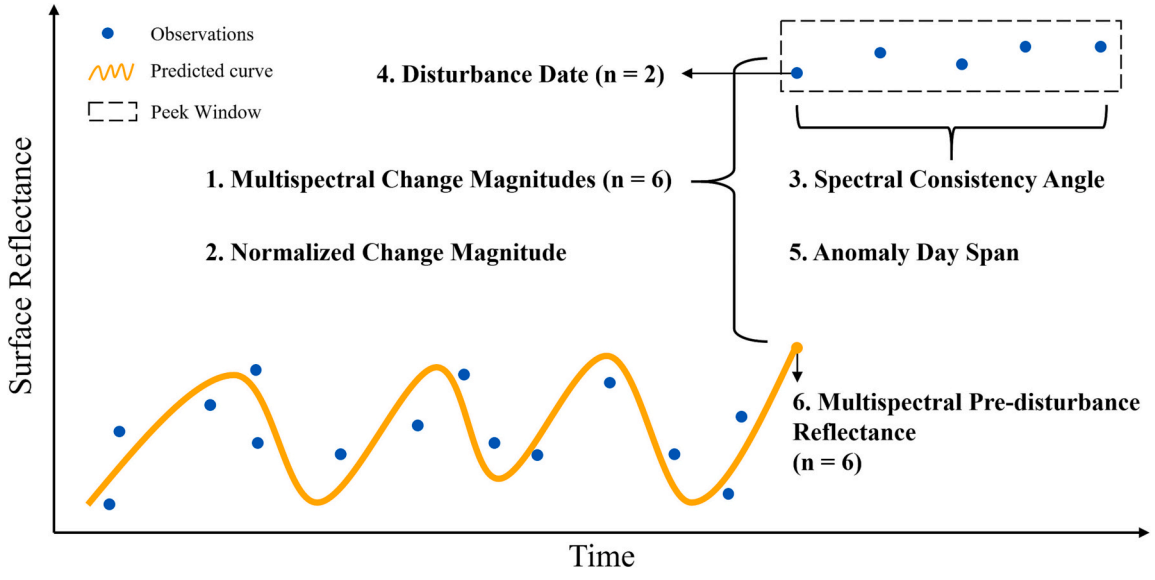


Fig. 4. Graphic illustration of the disturbance-related predictors used for this study ("n" means the number of variables).

Table 2
Definition of 17 predictors used for the NRT disturbance prediction.

Predictor name	Description	Number	Category
1. Multispectral change Magnitudes	$CM_i = \rho_{i,dist_dat} - \widehat{\rho}_{i,dist_dat},\ i = 1,2,3,4,5,6$	6	Disturbance process
2. Normalized change magnitude	$CM_{normalized} = \sum_{i=1}^{5}\left(\frac{CM_i}{RMSE_i}\right)^2$	1	
3. Spectral consistency angle	$angle = \frac{1}{n-1}\sum_{j=1}^{n-1}\beta_{j,j+1}$	1	
4. Disturbance date	$sin_{doy} = sin\left(\frac{2\pi}{T}doy_{dist_dat}\right)$, $cos_{doy} = cos\left(\frac{2\pi}{T}doy_{dist_dat}\right)$	2	
5. Anomaly span	$ndays = lastdist_dat - dist_dat$	1	
6. Multispectral pre-disturbance reflectance	$\widehat{\rho}_{i,dist_dat} = a_{0,i} + \sum_{k=1}^{3}\left(a_{k,i}cos\left(\frac{2\pi}{T}x\right) + b_{k,i}sin\left(\frac{2\pi}{T}x\right)\right) + c_{1,i}x,\ i = 1,2,3,4,5,6$	6	Pre-disturbance

Note: CM_i and $RMSE_i$ are the individual change magnitude and Root Mean Square Error of the ith band; $\rho_{i,dist_dat}$ and $\widehat{\rho}_{i,dist_dat}$ are the actual and the predicted reflectance of the ith band; n is the current consecutive anomaly number; $\beta_{j,j+1}$ is the change-vector angle from jth to $j+1$th anomaly; *lastdist_dat* and *dist_dat* are the last disturbance date observed as a spectral anomaly and the disturbance starting date respectively; doy is the date of the year; $a_{0,i}$, $a_{k,i}$, $b_{k,i}$ and $c_{1,i}$ are the harmonic coefficients for the ith band; k is the temporal frequency of the harmonic components.

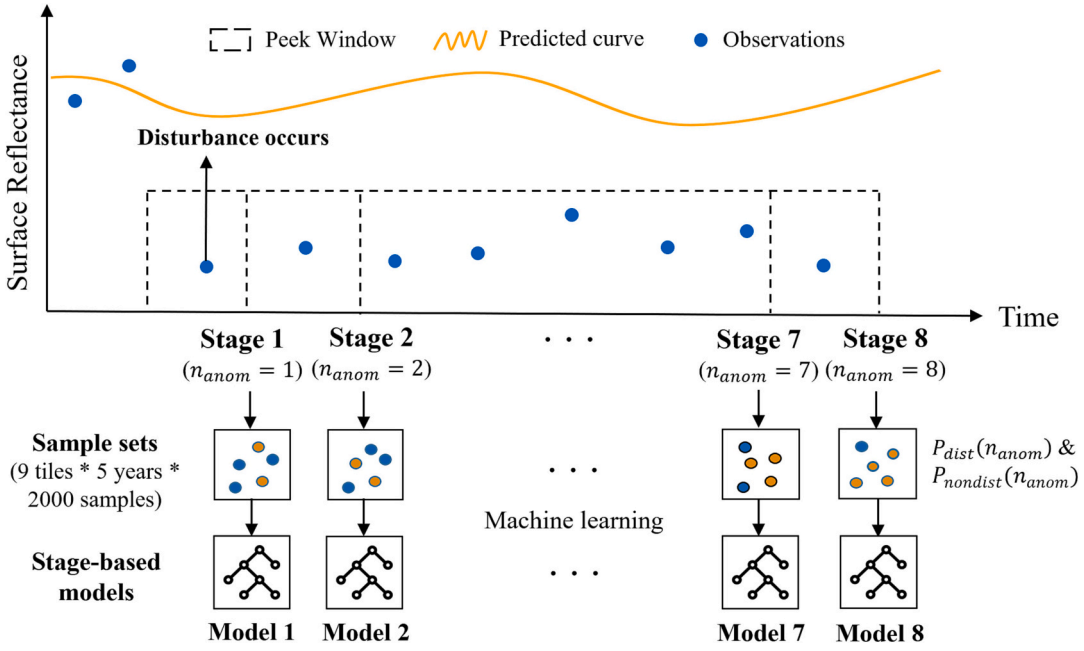


Fig. 5. Explanation for developing stage-based sample sets and disturbance models. The disturbance stages were characterized by the number of current consecutive anomalies (n_{anom}). We used the stage-specific sample proportion, $P_{dist}(n_{anom})$ and $P_{nondist}(n_{anom})$, to make samples and then developed a machine-learning model individually for each stage.

disturbances) using each corresponding sample set. Hence, a total of eight stage-based retrospective models were prepared for the second phase.

3.2. Stage 2: near real-time monitoring

Starting from January 1st, 2022, we iterated a workflow of near real-time monitoring to process the newly collected images until the end of 2022. The workflow was operated incrementally at a step of one week. The workflow ingested new images collected for each week, updated the NRT anomaly records, and then extracted 17 disturbance-related predictors listed in Table 2. We tested and compared two ways for generating disturbance probability: (1) the proposed method based on the retrospective models (called supervised) and (2) the traditional anomaly method based on an empirical change-magnitude threshold (called unsupervised). For the supervised probability, we first used the change probability threshold of 0.9 to detect anomalies for each pixel (the same *chg_t* for "anomaly generation" in Phase 2). Then, 17 predictors were generated for each outputted anomaly. Finally, we input the predictors into the corresponding stage-based model based on the current n_{anom}, and produced its NRT random-forest disturbance probability at a per-pixel level. We assigned the disturbance probability as "0" for those pixels without anomalies being detected. The unsupervised probability applied the change probability threshold of 0.99 to detect disturbance breakpoints, followed by eliminating the breakpoints attributed to the greener trend using the rule of Eq. (1). The threshold as 0.99 is the most commonly used change threshold for the CCDC-like approach, which, for example, was adopted in the "Land Change Monitoring, Assessment, and Projection" project (Brown et al., 2019). We computed the unsupervised probability as the ratio of the current consecutive anomaly number by the maximum anomaly number ($conse = 8$). For example, if there are six anomalies at the end of the current time series, the disturbance probability is 6 / 8 = 75%.

To generate spatially complete disturbance patches (i.e., a

continuous region affected by the change process), the Flood Fill algorithm was first applied to the two disturbance probability layers (Ye et al., 2023). Flood Fill is an algorithm to identify adjacent pixels as an object in an image based on their similarity to an initial seed pixel (Van der Walt et al., 2014). The Flood Fill algorithm outperformed other segmentation algorithms such as SLIC and watershed in our previous test (Ye et al., 2023), and hence was chosen for this study. For the supervised probability layer, we retained the disturbance regions as the disturbance patches with an average disturbance probability >50%; for the unsupervised-probability layer, we labeled the patches as their average probability ≥ 12.5%, so that the disturbance identification was possible even when one anomaly observation was captured, as consistently as the supervised-probability approach.

Finally, the identified disturbance patches were categorized into two levels of the NRT products based on the current patch-level anomaly number. We assigned the disturbance patches with an average of less than six consecutive anomalies into "detection level", and otherwise "confirmation level". We used the confirmation threshold as six consecutive anomalies, considering most CCDC-like algorithms use six observations as the minimum required anomaly number (Brown et al., 2019; Ye et al., 2021a; Zhu et al., 2020). When one workflow was completed for a monitoring interval, we saved two levels of the disturbance products with the disturbance attributes into the database. The polygon-based attributes include the first anomaly date, the last anomaly date, and the production date (we set it as the last day of each week). As our experiments were performed in early 2023, all HLS datasets for 2022 have been available for downloading, and hence the data lag was zero. To keep close to the real latency for the operational monitoring, we manually set the data lag as two days of HLS data lag and added it to all production dates. The two-day data lag is the estimated average time required by NASA LP DAAC to release HLS2.0 surface reflectance products since their first collection, based on the user experience from our NRT platform downloading over 90,000 HLS 2.0 images. It is worth mentioning that our comparative evaluation mainly focused on the detection-level products, though both the detection- and confirmation-level results were assessed and reported. This was because the detection level was practically more relevant for an NRT platform owing to its enabling timelier monitoring (<10 days) and providing important potential disturbance maps for further high-resolution applications. We developed an operational platform to update and to visualize the NRT results for each week based on the Google Earth Engine platform (https://gers.users.earthengine.app/view/nrt-conus).

3.3. Accuracy and latency assessment

We will use the sigmoid framework proposed by Bullock et al. (2022) to comparatively assess the unsupervised and supervised approach. The sigmoid framework incorporates both omission and commission of disturbance alerts as a function of lag. Hence, it could lead to a more comprehensive assessment than other NRT assessment metrics that only focused on the disturbance cases being correctly identified (Bullock et al., 2022), such as Mean Time Lag (*MTL*) (Reiche et al., 2018). In this study, we defined the omission error at a certain lag as when a disturbance sample was not hit by the disturbance patches from the recent NRT results and defined the commission error as when a non-disturbance sample was included in one disturbance patch.

Following the sigmoid framework, we will graph omission, commission, and F_1 score as sigmoid curves for incremental values of the total lag (i.e., production date minus disturbance date), allowing for a thorough comparison of detection effectiveness at a full range of latency. F_1 score is a harmonic mean of inverse omission and commission error rate:

$$F_1 = 2^{*}\frac{(1-omission)^{*}(1-commission)}{(2-omission-comission)} \quad (4)$$

Two lag metrics generated from the sigmoid curves will be used as the primary tool to assess "lag-accuracy" performance: the Initial Delay and Level Off Point. The Initial Delay is assigned to the date that 2% of the total disturbance pixels are detected, representing the shortest time of an algorithm for beginning to alert a disturbance event. The Level Off Point, the time when F_1 score is just stabilized, is computed as the date that has the maximum y for a virtual curve from rotating 45 degrees around the line connecting the beginning and end points of the sigmoid curve (Bullock et al., 2022). The x coordinate of Level Off Point indicated the shortest lag to reach the maximum F_1 score, and the y coordinate points to the maximum overall performance.

Moreover, we will assess the individual contribution of the five lag sources given in Fig. 1. We will generate individual lags for each disturbance sample following the definitions given in Table 3, and then compute the mean statistics of each lag component. We will also report Mean Time Lag (*MTL*) and Mean Time Lag for the First Alert (MTL_f) for comparison to other existing studies. The sum of all the lag sources is equivalent to *MTL*, a lag metric commonly used in previous studies (Shang et al., 2022; Ye et al., 2021a). The sum of all except the confirmation lag is equal to as MTL_f (Reiche et al., 2018), which is the average time between the reference disturbance date and the date for capturing the first anomaly.

4. Results

4.1. Unsupervised vs. supervised

Fig. 6 shows the sigmoid curves for the unsupervised (the traditional method based on a predefined threshold) and the supervised (the proposed) NRT approach based upon all evaluation samples from the five testing sites ($n = 3000$, 50% are disturbance samples). Table 4 lists their key lag metrics. Generally, F_1 score curves for the two approaches both began to increase around the Initial Delay (Fig. 6A); with time going forward, F_1 was gradually rising because more post-disturbance observations became available both temporally and spatially, and eventually plateaued at the Level Off Point (Fig. 6A). The sigmoid curves of the two approaches were almost fully overlapped before their Initial Delay (2 days for the detection level, and ~ 21 days for the confirmation level), and became divergent afterward: the F_1 sigmoid curve of the supervised approach ("blue") became consistently higher than that of the unsupervised approach ("yellow"). Particularly at the detection level, the supervised approach only took 15 days to reach the Level Off Point with 0.733 F_1 score, which had seven days quicker and almost 0.2 F_1 improvement than the unsupervised (the unsupervised: 22 days, 0.546 F_1 score, see Table 4). The detection-level comparison reflects their large performance distinction for mapping disturbances at their early stage. Differently, the unsupervised and supervised approaches presented the same lag days to reach their Level Off Points for the confirmation level (35 days), because the two approaches both applied *conse* = 6 hence similar confirmation lag (as Section 4.2 will show) that dominates their

Table 3
Definition of lag types to be assessed for this study. *MTL*: Mean Time Lag; MTL_f: Mean Time Lag for the First Alert.

Lag types	Definition
Observation Lag	Day interval between the reference disturbance date and the first clear observation date
Data Lag	Constant value as two days
Sensitivity Lag	Day interval between the first clear observation date and the first anomaly date detected
Confirmation Lag	Day interval between the first and the last anomaly date detected for a peek window
Production Lag	Day interval between the last anomaly date and the date being officially confirmed by the platform
MTL	The sum for observation lag, data lag, sensitivity lag, confirmation lag, and production Lag
MTL_f	The sum for observation lag, data lag, sensitivity lag, and production Lag

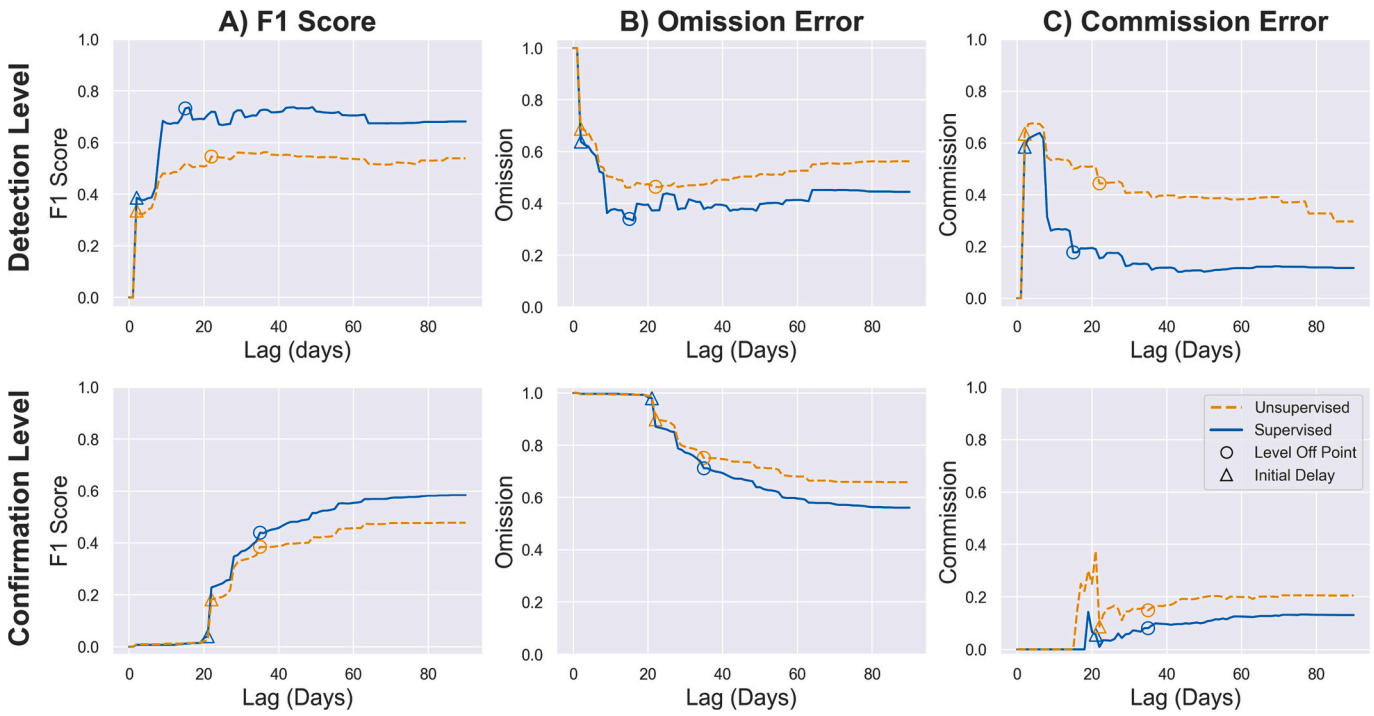


Fig. 6. Comparison of the unsupervised and supervised approaches on sigmoid curves at detection and confirmation levels based on a total of 3000 evaluation samples from the five disturbance sites. The omission error, commission error, and F_1 score are respectively plotted as a function of total lag (the sum of five lag components) within a tolerance window of 0 to 90 days. Two key latency metrics, Initial Delay and Level Off Points, are marked on the curves as "circle" and "triangle".

Table 4
Comparison of the unsupervised and supervised approaches on lag, omission error, commission error and F_1 score at the Initial Delay and Level Off Point of their sigmoid curves. The bold font highlights the lag/accuracy difference between the two approaches.

	Detection Level							
	Initial Delay				Level Off Point			
	Lags (Days)	Omission Error	Commission Error	F_1	Lags (Days)	Omission Error	Commission Error	F_1
Unsupervised	2	68.8%	63.5%	0.336	**22**	46.4%	**44.4%**	**0.546**
Supervised	2	63.9%	58.5%	0.386	**15**	34.0%	**17.7%**	**0.733**
	Confirmation Level							
	Initial Delay				Level Off Point			
	Lags (Days)	Omission Error	Commission Error	F_1	Lags (Days)	Omission Error	Commission Error	F_1
Unsupervised	22	89.9%	8.7%	0.182	35	75.2%	14.9%	0.384
Supervised	21	98.0%	0.6%	0.040	35	71.2%	8.1%	0.439

total lag at the confirmation level (~77% of the total lag, see Section 4.2). Reducing consecutive observations for confirmation is out of the scope for this study and was often determined by user settings (Bullock et al., 2022). However, the supervised approach still achieved a relatively better F_1 score at the confirmation level (supervised vs. unsupervised: 0.439 vs. 0.384).

By investigating the error components, the supervised presented consistently lower omission (Fig. 6B) and commission rates (Fig. 6C) than the unsupervised approach over time after the Initial Delay. Particularly, the supervised achieved a much lower commission error rate than the unsupervised by approximately 27 percent points at the detection level (17.7% vs. 44.4%, Table 4). This revealed that the primary advantage of incorporating historical datasets was in alleviating overdetection. Fig. 7 showcased a sample pixel under the effects of climate variability that was misclassified as "disturbance" by the unsupervised approach. Its NIR time series experienced a noticeable increase at the breakpoint (Fig. 7A), which was related to the earlier leaf-on date for the year 2022 (Fig. 7B). The unsupervised approach identified the breakpoint as "disturbance" because the overall change magnitude was larger than the predefined threshold (the change probability of 0.99), while the supervised correctly ignored it (Fig. 7C) through a comprehensive examination on change magnitudes, pre-change features, change season, and change direction using the retrospective models.

Fig. 8 is an example to show that the supervised approach also could suppress omission errors. The sample pixel was experiencing spongy moth damage with a medium-level NDVI drop in June of 2022 (Fig. 8A and B). Unless a structural breakpoint was detected so that the S-CCD model entered its initialization stage, the supervised approach inspected the outputted consecutive spectral anomalies (if any) and predicted disturbance probability for each new observation. Therefore, the supervised approach was capable of delineating subtle change induced by spongy moth (Fig. 8C) even with low change magnitudes, as long as most pixels within the associated anomaly patch were identified as disturbance pixels by the machine-learning model. However, the unsupervised approach missed most damage at the confirmation level (Fig. 8C) as the spectral curve was quickly recovered (Fig. 8A), causing the change magnitude to be lower than the threshold of 0.99 change probability.

Comparing the detection and confirmation levels, the accuracies at the detection level were higher than at the confirmation level for both

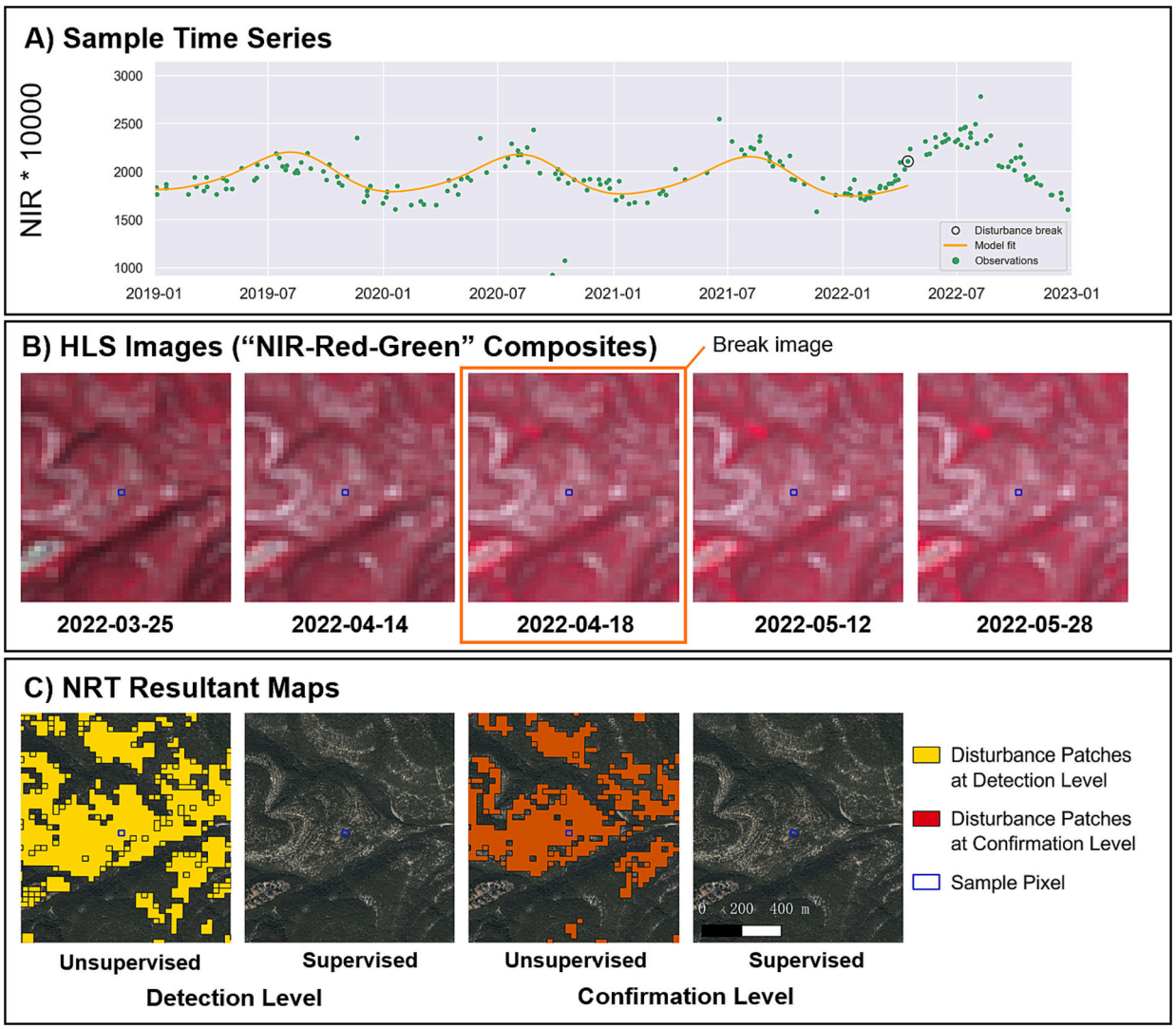


Fig. 7. An example site (−97.947, 30.339) for commission errors produced by the unsupervised approach. The HLS time series (A) shows a structural breakpoint induced by increasing NIR which was possibly related to phenological shifts, while no obvious disturbance signal was found from inspecting the HLS images (B). The supervised approach exhibited superiority in suppressing such over-detection by modeling disturbance and non-disturbance anomalies using the retrospective datasets (C).

approaches. This finding was contrary to most past NRT studies, in which the confirmation results achieved the better overall accuracies as only enough consecutive anomalies could effectively exclude commission errors. The outstanding performance for the detection level was mainly attributable to its considerably lower omission errors (e.g., detection vs. confirmation = 34.0% vs. 71.2% for the supervised at its Level Off Point). Notably, the detection level was advantageous for capturing the ephemeral disturbances (e.g., Hurricane Ian and Spongy Moth) whose effects only lasted for one or two observations. As two out of five study sites were ephemeral disturbances (344 / 830 disturbance cases), the validation samples for this disturbance type might be over-represented for this study, resulting in an overestimation of the detection-level accuracy. Note that the unrealistic sample proportion only distorted the relative accuracy between the detection and the confirmation level but yielded little statistical effects on comparative performance between the unsupervised and supervised approaches. Another distinction between the detection and the confirmation level was that the confirmation level required a much longer lag to reach Initial Delay (22/21 days vs. 2 days) and Level Off Point (35 days vs. 22/15 days) than the detection level for both unsupervised and supervised approaches. Their difference reflects the minimum lag days (roughly 20 days) required to collect six HLS clear observations after disturbance occurrence.

4.2. Lag component evaluation

Fig. 9 displays assessments of different lag components of the two approaches. Note that we made assessments based on only the disturbance samples that were successfully confirmed by each approach (unsupervised: 287 cases; supervised: 370 cases). The total lags of the two approaches, *MTLs*, are close (unsupervised vs. supervised = 39.4 vs. 38.8 days), which is mainly due to 1) the ambiguity in reporting independent temporal metrics with ignorance on spatial accuracy (i.e., omission and commission error rates) (Bullock et al., 2022); 2) the total lags of the two approaches were dominated by their similar confirmation lag (~77%) as their required anomaly numbers are both set as six. In comparison, the supervised approach yielded lower sensitivity lags (0.1 vs 1.1 days), making its MTL_f shorter than the unsupervised by 10% (9.1 vs. 10.0 days). This revealed the advantage of the supervised approach for the early alerting of the disturbance, especially when the signal was not pronounced. The observation lags were both averagely 2.0 days, which were mainly controlled only by the HLS revisiting cycle and

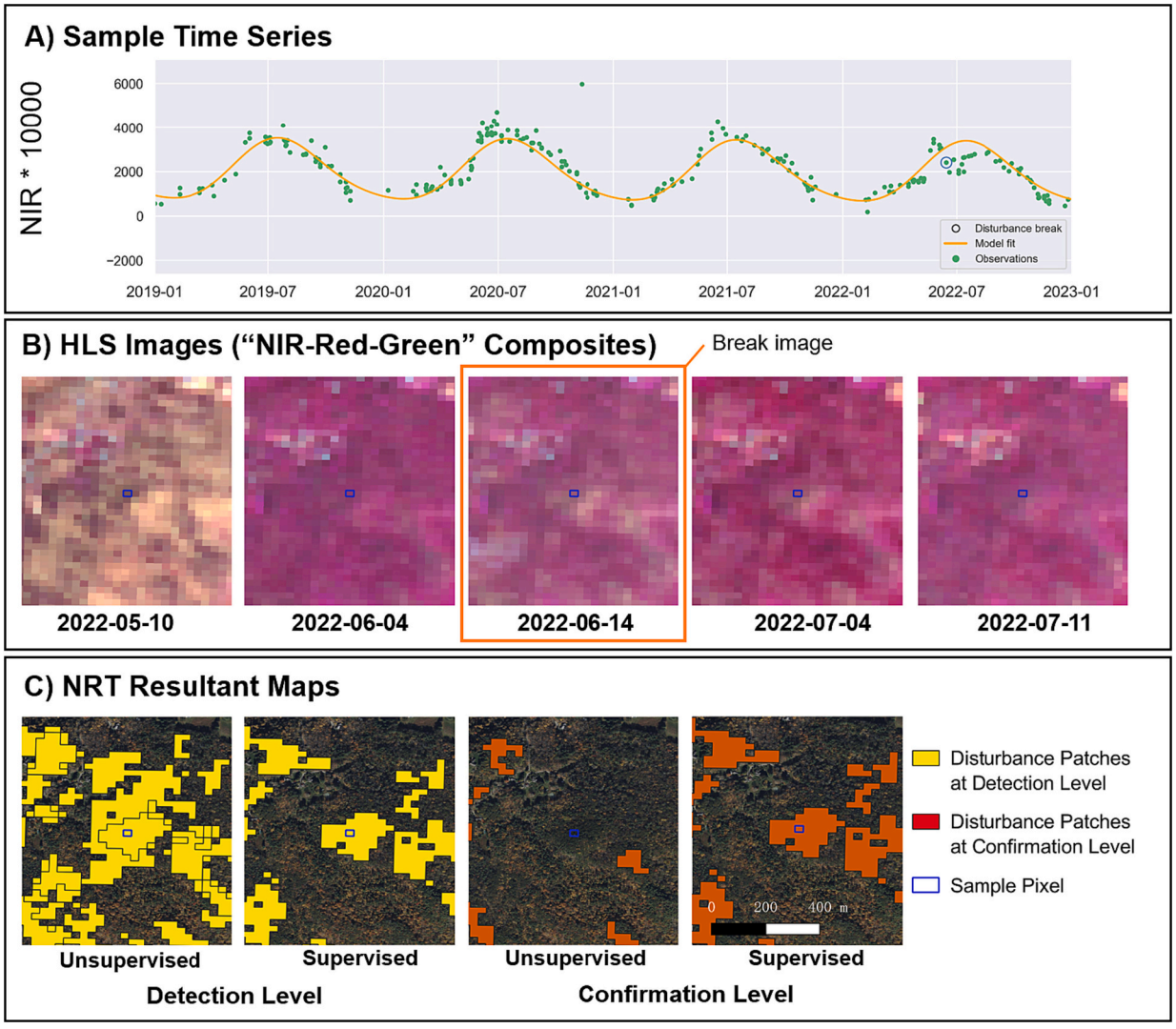


Fig. 8. An example site (−73.377, 41.946) for omission errors produced by the unsupervised approach. The HLS time series (A) shows a time series impacted by spongy moth damage, with the disturbance breakpoint detected by the supervised approach marked as "black circle" in (A). The damage has been verified by color change ("white spots") within HLS images (B). The NRT results (C) present that the unsupervised approach totally missed the disturbance patches at the confirmation level due to the quick spectral recovery.

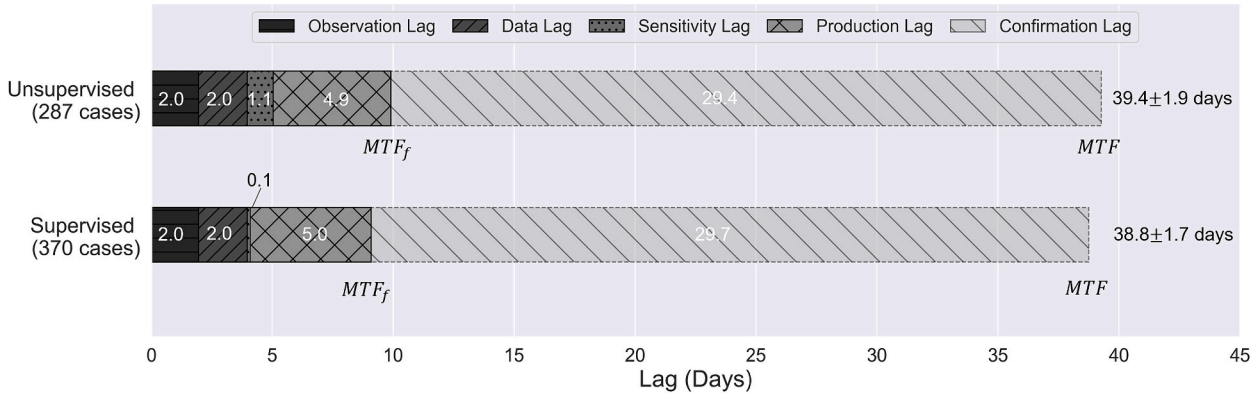


Fig. 9. An assessment of the unsupervised and supervised approach for their different lag components, with their correctly confirmed sample case number shown in the y-axis labels. *MTL*: Mean Time Lag; MTL_f: Mean Time Lag for the First Alert.

weather conditions. The average production lags of the two approaches were almost the same (4.9 vs. 5.0 days). Their small numeric difference was caused by the different sample sets that were successfully confirmed (unsupervised vs. supervised = 287: 370 cases) and involved in production lag assessments. *MTL* and MTL_f of the two approaches did not present sufficient differences as their sigmoid curves showed (Fig. 6),

which echoed the viewpoint that these two metrics are problematic for latency assessment because they were calculated based on their own correctly identified samples, ignoring their omission and commission errors (Bullock et al., 2022). A high rate of omission errors and commission errors often caused incomplete patch detection and too many noisy signals, which would eventually lead to a mapping delay.

4.3. Individual disturbance assessments

Fig. 10 shows that individual disturbance types presented different sigmoid curves and lag metrics. For the detection level, "Kentucky flood" (Fig. 10C) received the quickest alerting as its Level Off Point at (9 days, 0.94 F_1 score); "Construction" had the longest latency as the Level Off Point at (43 days, 0.68 F_1 score), which was possibly due to the fact that the initial stage of a construction project (e.g., land clearing) often yielded small-area and unobvious spectral signals; two ephemeral disturbances, "Kentucky Flood" and "Hurricane Ian", presented an inverted U-shaped sigmoid curve while the other three were S-shaped. This indicates that some patches affected by ephemeral disturbances experienced rapid recovery. For the confirmation level, F_1 scores for two ephemeral disturbances dropped dramatically compared to the detection level, while the other three remained similar F_1 scores for their Level Off Point, which explained the noticeable performance difference between the detection and the confirmation level in Table 4.

Fig. 11 demonstrates the dynamics of the detection- and the confirmation-level products over time since a disturbance. For "Mosquito Fire" (Fig. 11A) and "Spongy Moth" (Fig. 11B), most disturbance regions in the detection level were confirmed (displayed as red polygons) since 5th week in the figure. Differently, the detected patches for two ephemeral disturbances, "Hurricane Ian" (Fig. 11C) and "Kentucky Flood" (Fig. 11D), were not converted into the confirmation-level patches but disappeared after the 5th week, as these disturbance events did not last long enough to generate six consistent anomalies. The NRT platform did not successfully map the Kentucky Flood in the 1st week for this case, which might be lacking clear observations due to the rainy/cloudy weather often accompanying the flooding event (longer data lag); once the first post-flood clear observation became available, more flood-related signals were captured, mainly on housing damage and soil change, leading to an F_1 jump at the 9th day in Fig. 10C ("detection level").

Table 5 exhibits different algorithmic latency (sensitivity and confirmation lags) among the five disturbance events. The other three lag components (data, observation, and product lags) were only contingent on data acquisition or platform operation, thereby generally indifferent across various disturbance events. As a result of the subtle change signals at their early stage, spongy moth and construction yielded higher sensitivity lags for the NRT monitoring. Compared to the unsupervised, the supervised approach reached a lower average sensitivity lag, enabling an earlier alerting for these two events. For the confirmation lag, Mosquito fire showed much shorter lags than the other four disturbances, possibly owing to the frequent dry and sunny weather for North California providing dense post-fire observations. Different from the sensitivity lag, confirmation lags were close between the supervised and the unsupervised approaches, indicating that the retrospective information helped little in reducing the confirmation lag, which should be controlled by the setting of *conse* and data density.

5. Discussion

5.1. Comparison to other NRT studies

Compared to the study using the closest evaluation method (Bullock et al., 2022), applying CCDC on sentinel-1 time series (i.e., "P2" configuration) reached a Level Off Point (203 days, 0.216 F_1 score) at the confirmation level for monitoring tropical deforestation, much worse than the same metric result for the supervised approach (38.8 days, 0.439 F_1 score). Not to mention that we included the ephemeral disturbances which were challenging and greatly impacted our F_1 score at the confirmation level. Reiche et al. (2018) combined observations from six satellites, Sentinel-1, ALOS-2, Landsat 7 and 8, to detect a specific land disturbance, deforestation, and reached a *MTL* as 31 days; Shang et al. (2022) built a method based on HLS datasets with setting four minimum consecutive anomalies for confirmation, reporting a resultant *MTL* as 35 days. The *MTL* of our supervised approach (38.8 days) seems to be a little longer than the above two studies, but we approached more diversified land disturbances including ephemeral disturbances. More important is that we assessed our approach in an operational platform by including observation, data, and production lag assessments, which additionally increased approximately nine extra lag days compared to the previous studies. It is noteworthy that for emergency events, the updating interval of our NRT system could be manually adjusted to a one-day interval to make the fastest response with

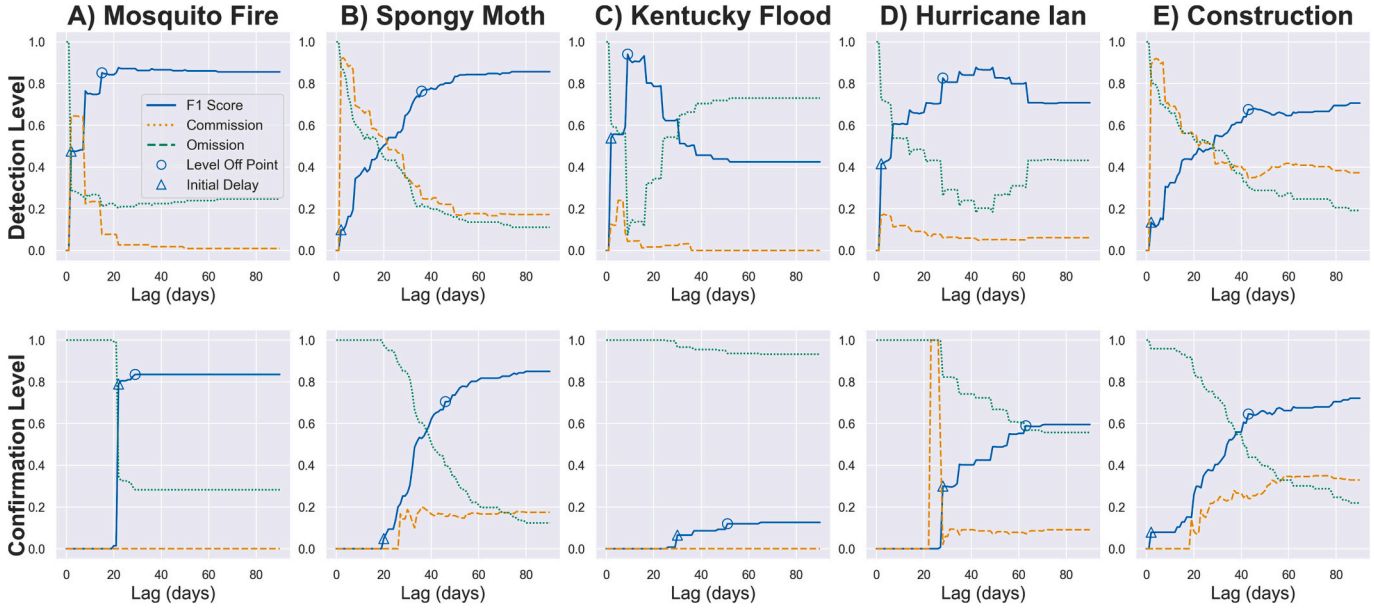


Fig. 10. Sigmoid accuracy curves for individual disturbance events being marked with the two key latency metrics at the detection and at the confirmation level.

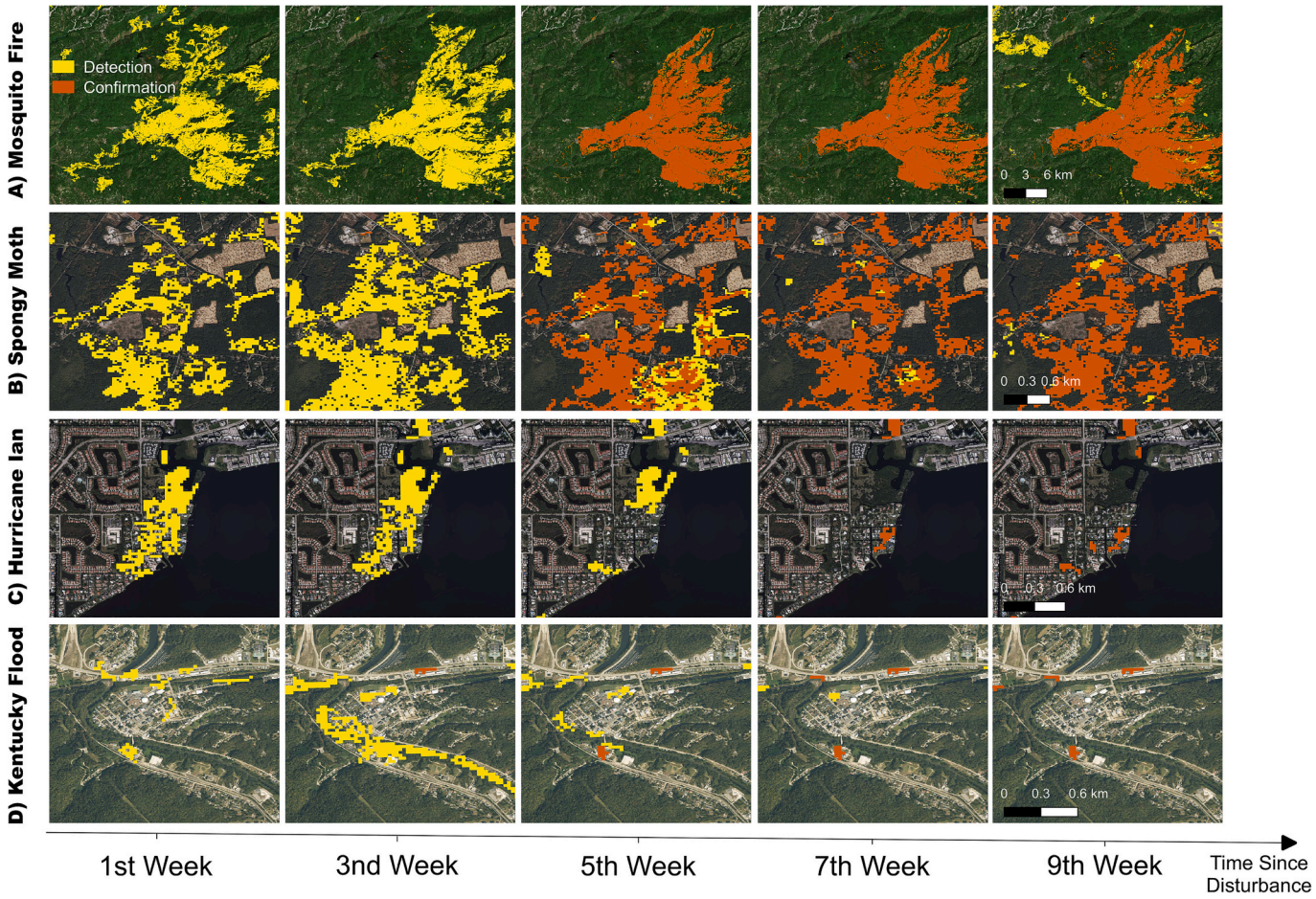


Fig. 11. The dynamics of detection- and confirmation-level products for different disturbance events. The centroid coordinates for the sites are Mosquito Fire (−120.752, 39.056), Spongy Moth (−73.684, 43.144), Hurricane Ian (−81.898, 26.652), and Kentucky Flood (−83.382, 37.553).

Table 5
Algorithmic latency comparison for the five disturbance events.

Sensitivity lag (days)					
	Mosquito Fire	Spongy Moth	Kentucky Flood	Hurricane Ian	Construction
Unsupervised	0.56	**4.29**	0	0	**6.40**
Supervised	0.57	**2.18**	0	0	**4.75**
Confirmation lag (days)					
	Mosquito Fire	Spongy Moth	Kentucky Flood	Hurricane Ian	Construction
Unsupervised	18.08	31.08	29.00	35.27	34.70
Supervised	17.22	31.80	32.89	35.36	35.13

zero-day production lag, so that the average lag for alerting the first anomaly for an event (MTL_f) would be further shortened to around half of a week (4.1 days). It will be even faster in the future, with more medium-resolution satellite sensors launched and quicker release of reflectance products.

All the previous studies used the unsupervised anomaly approach based on the predefined change threshold, mostly focusing on forest disturbances. To our knowledge, this study is the first attempt to build up a sole NRT approach/system intended for various natural disasters and human-induced terrestrial changes. To address the issue of disturbance variety, we designed multifaceted predictors, including the pre-disturbance features that are indicative of background land cover, and spectral change features for characterizing a disturbance process. With the retrospective analysis of those historical disturbances, we were able to establish machine learning models using these predictors to better analyze ongoing anomaly signals. Notably, our approach built an explicit model for each anomaly number condition, considerably enhancing the capability of distinguishing early signals and decreasing sensitivity lags, hence a timelier detection. For future research, the supervised approach is readily expanded to provide NRT mapping for disturbance agents through the incorporation of agent samples.

5.2. Limitations

The relatively lower overall accuracy in the confirmation level suggests the dilemma of disturbance confirmation. For this study, we used the CCDC default minimum observations (*conse* = 6) for confirming a disturbance patch, which was well verified in multiple past studies on Landsat-based studies (Brown et al., 2019; Ye et al., 2021a; Zhu et al., 2019). But the *conse* was not tested yet on a per-event basis. Especially, it was still challenging to accurately confirm the impacted regions of those short-lived events. The two ephemeral events, “Kentucky Flood” and

"Hurricane Ian", did not lead to permanent land cover conversion, and their spectral signals only lasted for one to three observations. As a result, their omission errors greatly increased after their Level Off Points (Fig. 10C and Fig. 10D). Even with a decreased minimum anomaly number such as *conse* = 4 in (Shang et al., 2022), it is still not likely to accurately detect events such as flash flooding, which often induces only one anomaly. In contrast, "Spongy Moth" (36 days, 0.76 F_1 score) and "Construction" (43 days, 0.68 F_1 score) had a lag for their Level Off Points close to the *MTL* of the supervised (38.8 days) using six anomalies. It suggests that confirming a disturbance event with six anomalies (*conse* = 6) is still workable for most non-ephemeral disturbance events. The varied sigmoid curves and Level Off Points of these disturbance events indicate a future direction for developing agent-based confirmation strategies instead of a universal confirmation rule.

The present methodology predominantly depends on U.S. disturbance datasets to construct retrospective models, which introduces difficulties in deploying the supervised approach globally. This could potentially be overcome by manual sample interpretation, but we propose further research into automating the transfer of U.S. samples to other countries with a goal of enhancing the global applicability of this supervised model. The method's efficacy is confirmed with the HLS dataset due to its high temporal density. However, there is a lack of clear observations in certain areas, such as tropical regions, or the disturbances associated with frequent cloud cover, like floods. For future improvements, we recommend integrating radar sensors like Sentinel-1.

6. Conclusion

We proposed a novel supervised machine learning approach to analyze spectral anomalies from HLS time series. The retrospective analysis extracted and modeled the early disturbance information from the historical time-series datasets, combined with existing disturbance products. A machine-learning model was built specifically for each anomaly number condition to better discriminate early disturbance signals. Compared to the unsupervised approach, the supervised approach reduced sensitivity lag, especially for subtle disturbances, and brought forward the Level Off point by seven days, with a 0.2 F_1 score improvement. Besides, the supervised approach considerably suppressed the commission error rate by 27 percent points, greatly enhancing the NRT capability for accurately delineating disturbance regions.

CRediT authorship contribution statement

Su Ye: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft. **Zhe Zhu:** Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Supervision, Writing – review & editing. **Ji Won Suh:** Data curation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2024.114071.

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